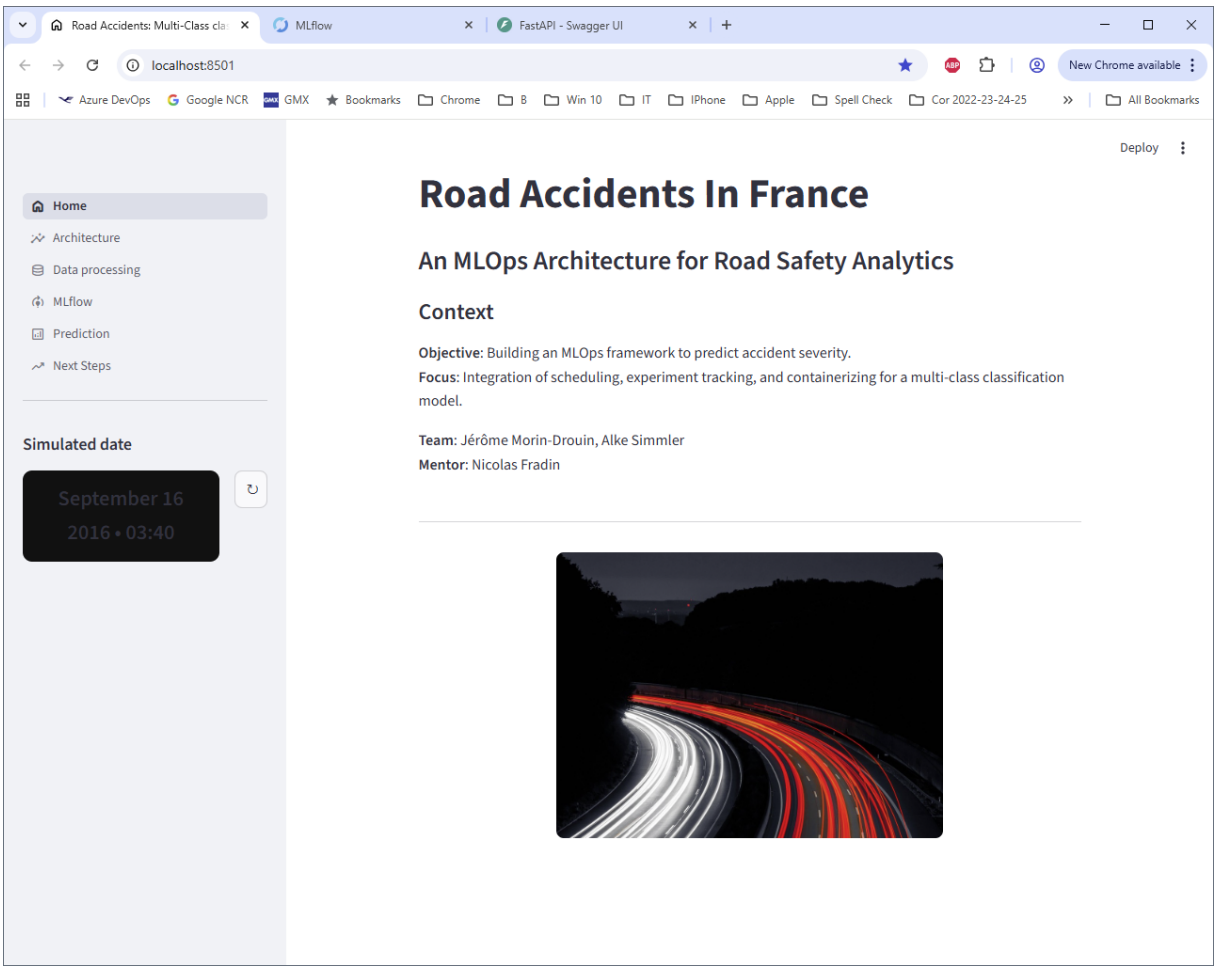
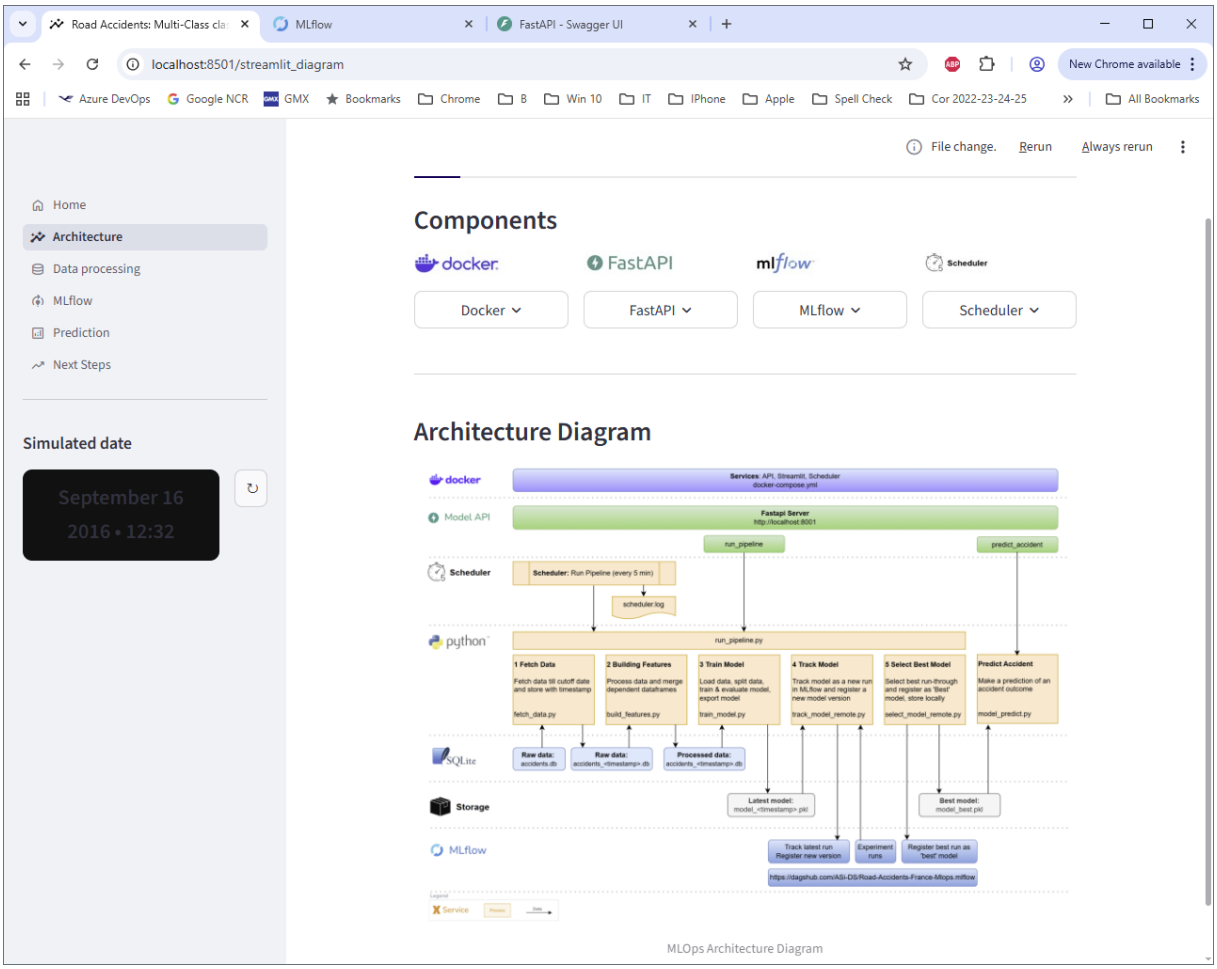
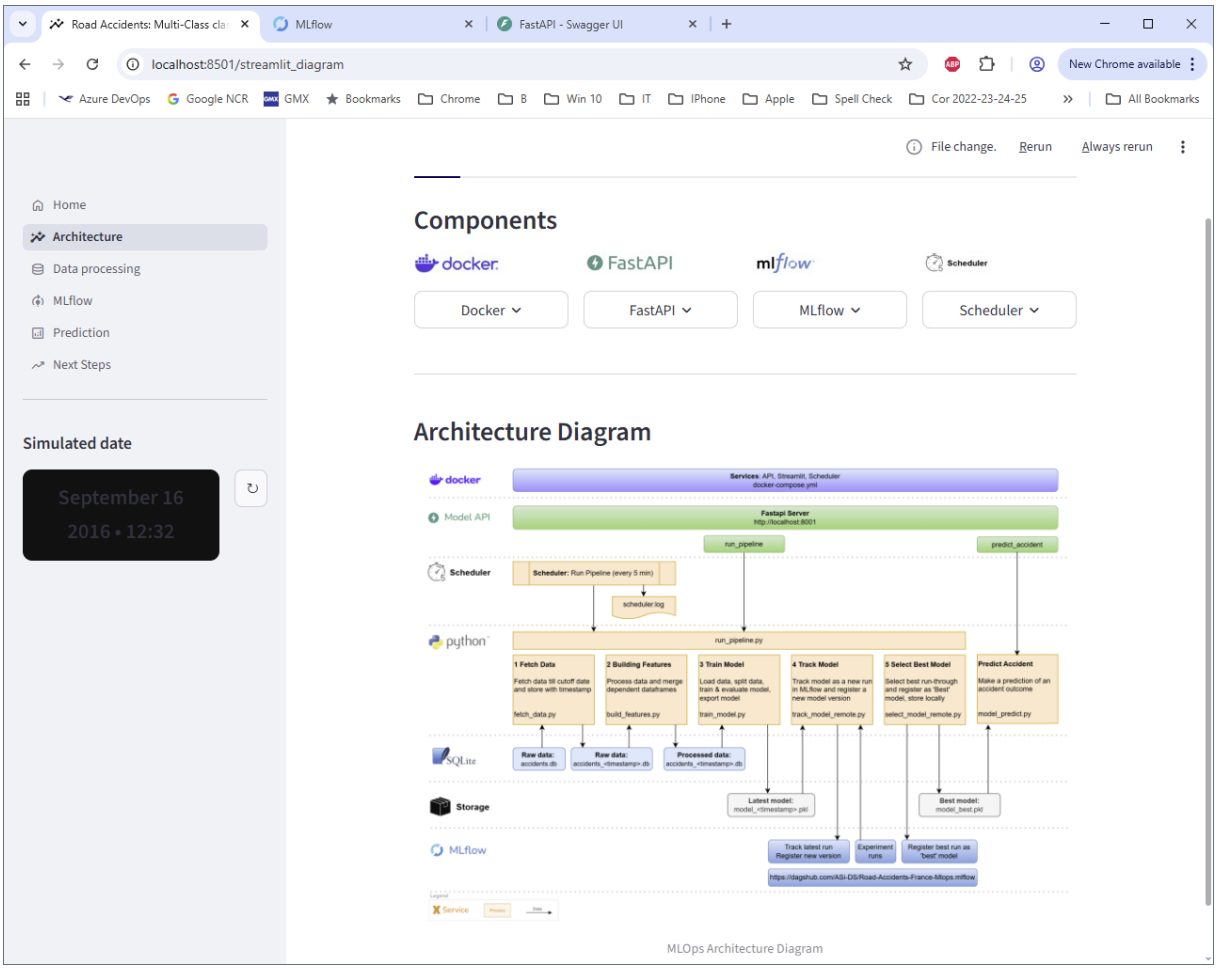
# Presentation: Road Accidents in France

Streamlit Screenshots

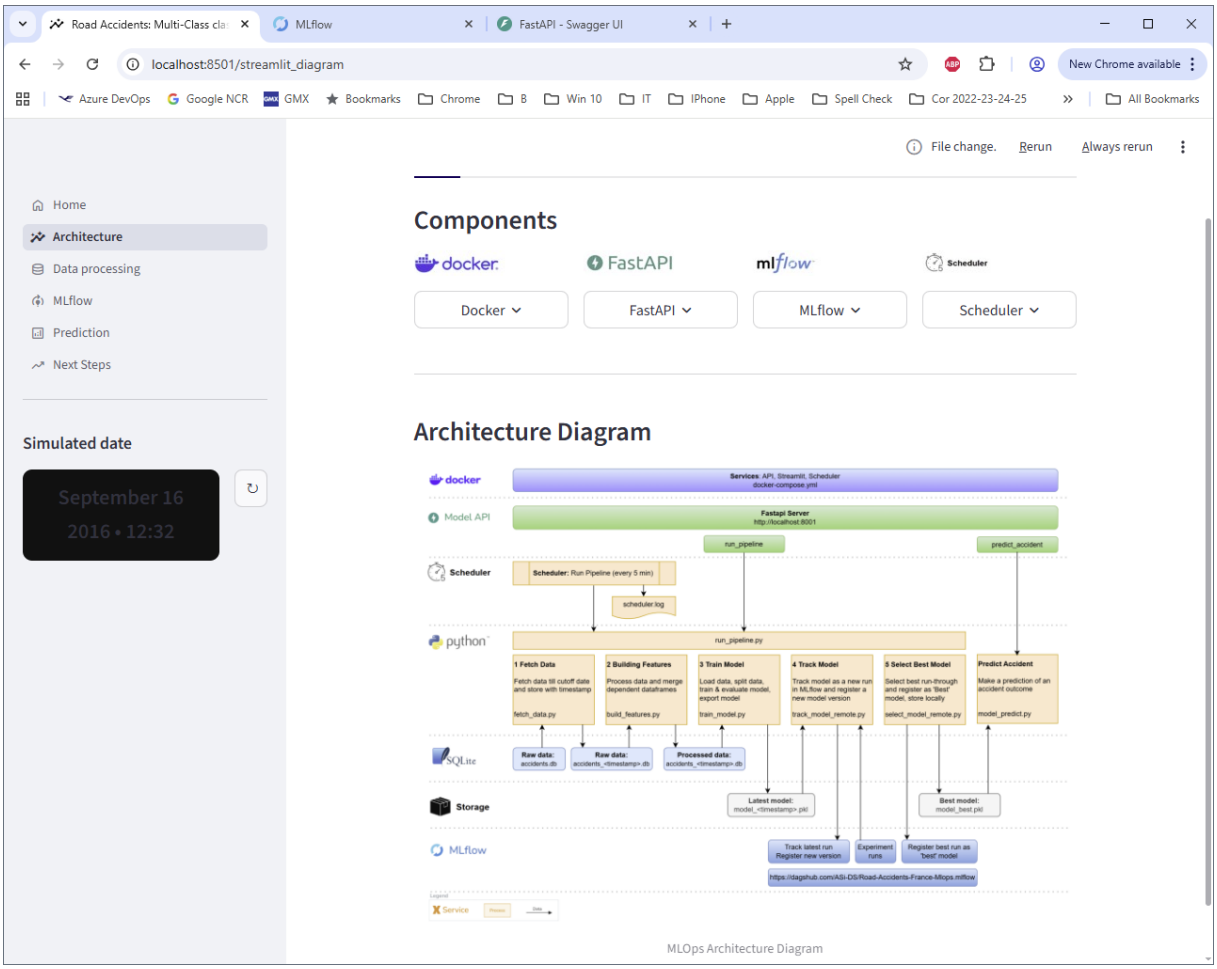




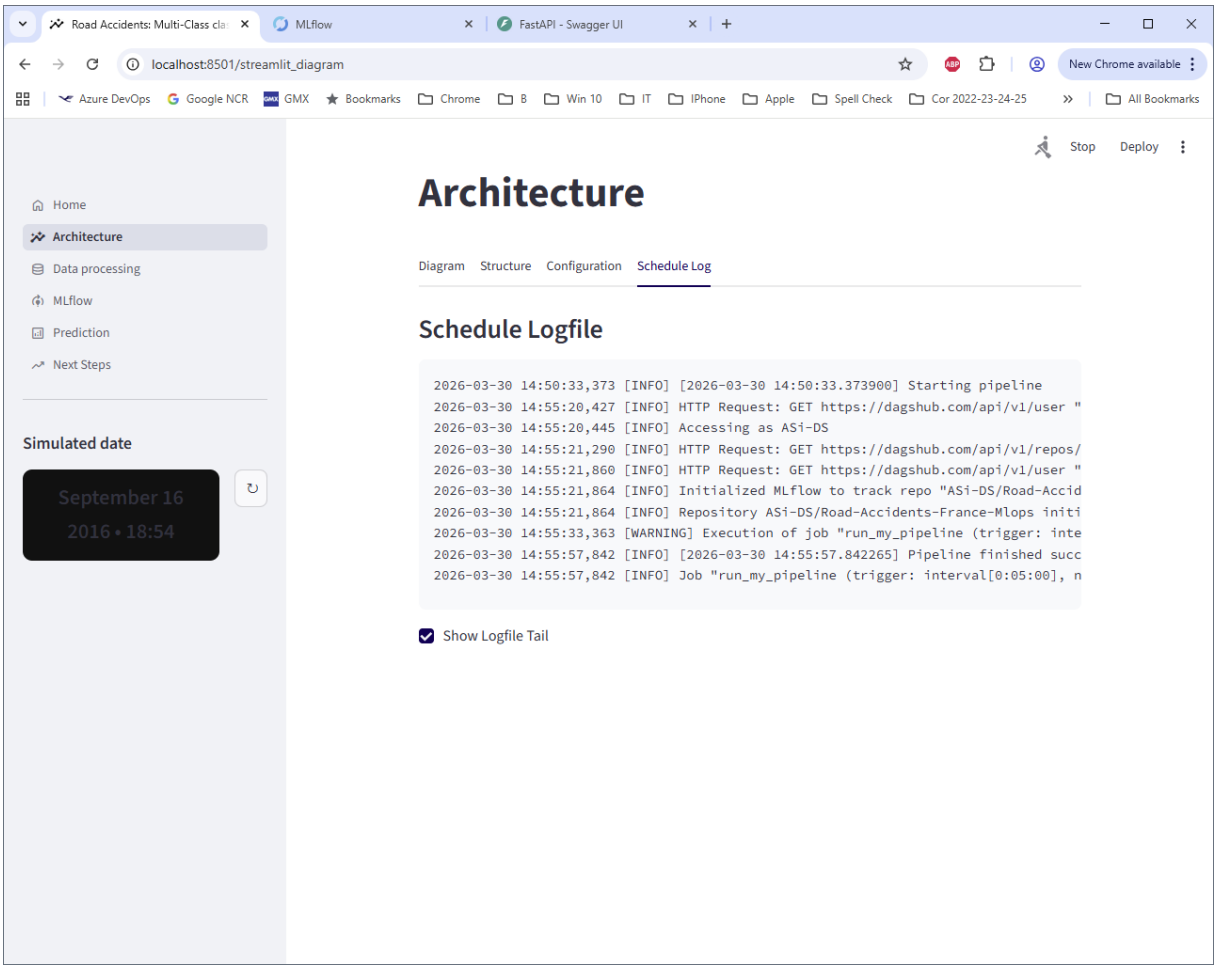

MLOps Architecture Diagram




MLOps Architecture Diagram




MLOps Architecture Diagram


Road Accidents: Multi-Class cla
MLflow
FastAPI - Swagger UI
localhost:8501/streamlit_diagram
New Chrome available
Azure DevOps
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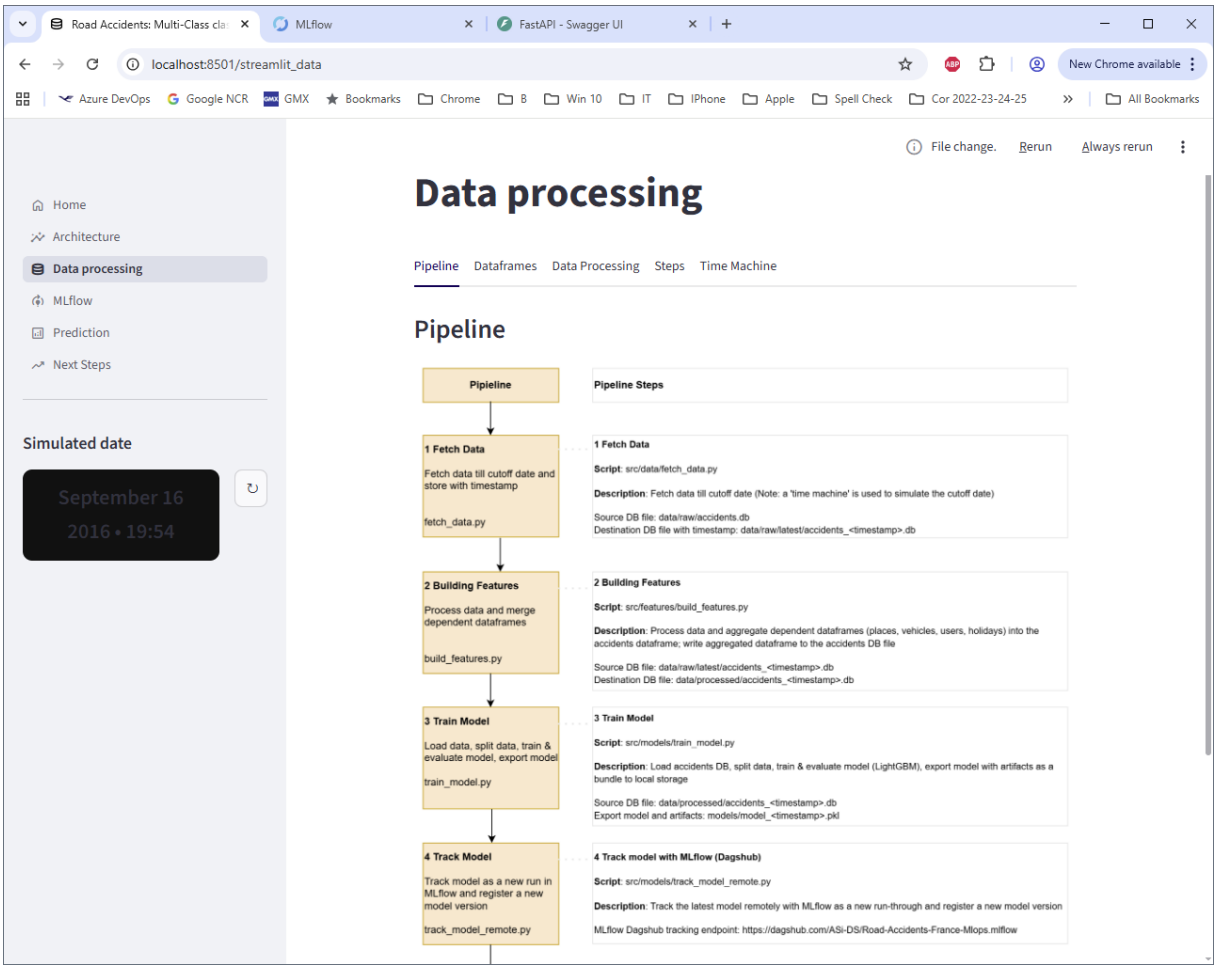
Structure
Configuration
Schedule Log
Schedule Logfile
2026-03-30 14:50:33,373 [INFO] [2026-03-30 14:50:33.373900] Starting pipeline
2026-03-30 14:55:20,427 [INFO] HTTP Request: GET https://dagshub.com/api/v1/user "
2026-03-30 14:55:20,445 [INFO] Accessing as ASi-DS
2026-03-30 14:55:21,290 [INFO] HTTP Request: GET https://dagshub.com/api/v1/repos/
2026-03-30 14:55:21,860 [INFO] HTTP Request: GET https://dagshub.com/api/v1/user "
2026-03-30 14:55:21,864 [INFO] Initialized MLflow to track repo "ASi-DS/Road-Accid
2026-03-30 14:55:21,864 [INFO] Repository ASi-DS/Road-Accidents-France-Mlops initi
2026-03-30 14:55:33,363 [WARNING] Execution of job "run_my_pipeline (trigger: inte
2026-03-30 14:55:57,842 [INFO] [2026-03-30 14:55:57.842265] Pipeline finished succ
2026-03-30 14:55:57,842 [INFO] Job "run_my_pipeline (trigger: interval[0:05:00], n
Show Logfile Tail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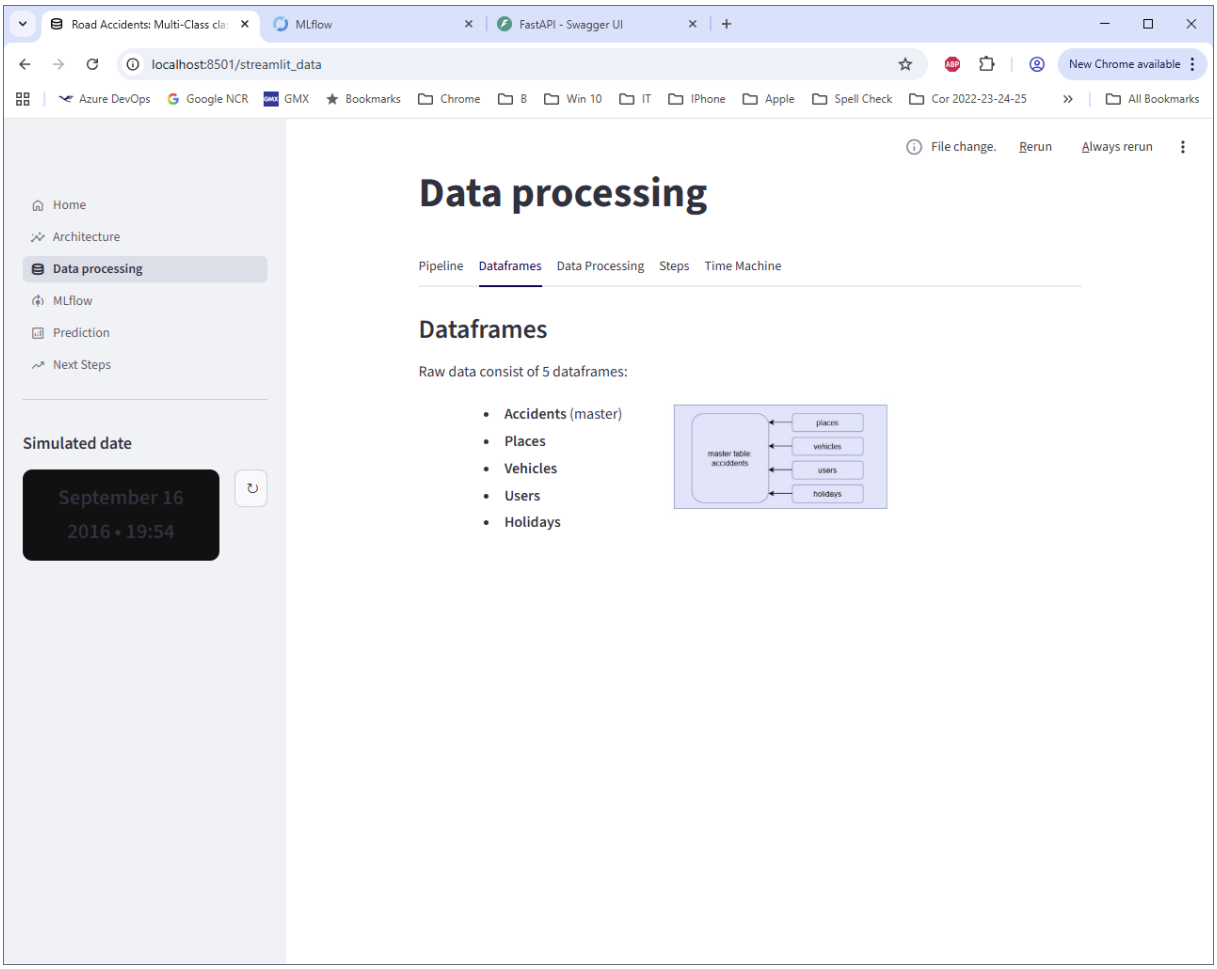
Pipeline Steps
1 Fetch Data
Fetch data till cutoff date and store with timestamp
fetch_data.py
1 Fetch Data
Script: src/data/fetch_data.py
Description: Fetch data till cutoff date (Note: a 'time machine' is used to simulate the cutoff date)
Source DB file: data/raw/accidents.db
Destination DB file with timestamp: data/raw/latest/accidents_<timestamp>.db
2 Building Features
Process data and merge dependent dataframes
build_features.py
2 Building Features
Script: src/features/build_features.py
Description: Process data and aggregate dependent dataframes (places, vehicles, users, holidays) into the accidents dataframe; write aggregated dataframe to the accidents DB file
Source DB file: data/raw/latest/accidents_<timestamp>.db
Destination DB file: data/processed/accidents_<timestamp>.db
3 Train Model
Load data, split data, train & evaluate model, export model
train_model.py
3 Train Model
Script: src/models/train_model.py
Description: Load accidents DB, split data, train & evaluate model (LightGBM), export model with artifacts as a bundle to local storage
Source DB file: data/processed/accidents_<timestamp>.db
Export model and artifacts: models/model_<timestamp>.pkl
4 Track Model
Track model as a new run in MLflow and register a new model version
track_model_remote.py
4 Track model with MLflow (Dagshub)
Script: src/models/track_model_remote.py
Description: Track the latest model remotely with MLflow as a new run-through and register a new model version
MLflow Dagshub tracking endpoint: https://dagshub.com/ASi-DS/Road-Accidents-France-Mlops.mlflow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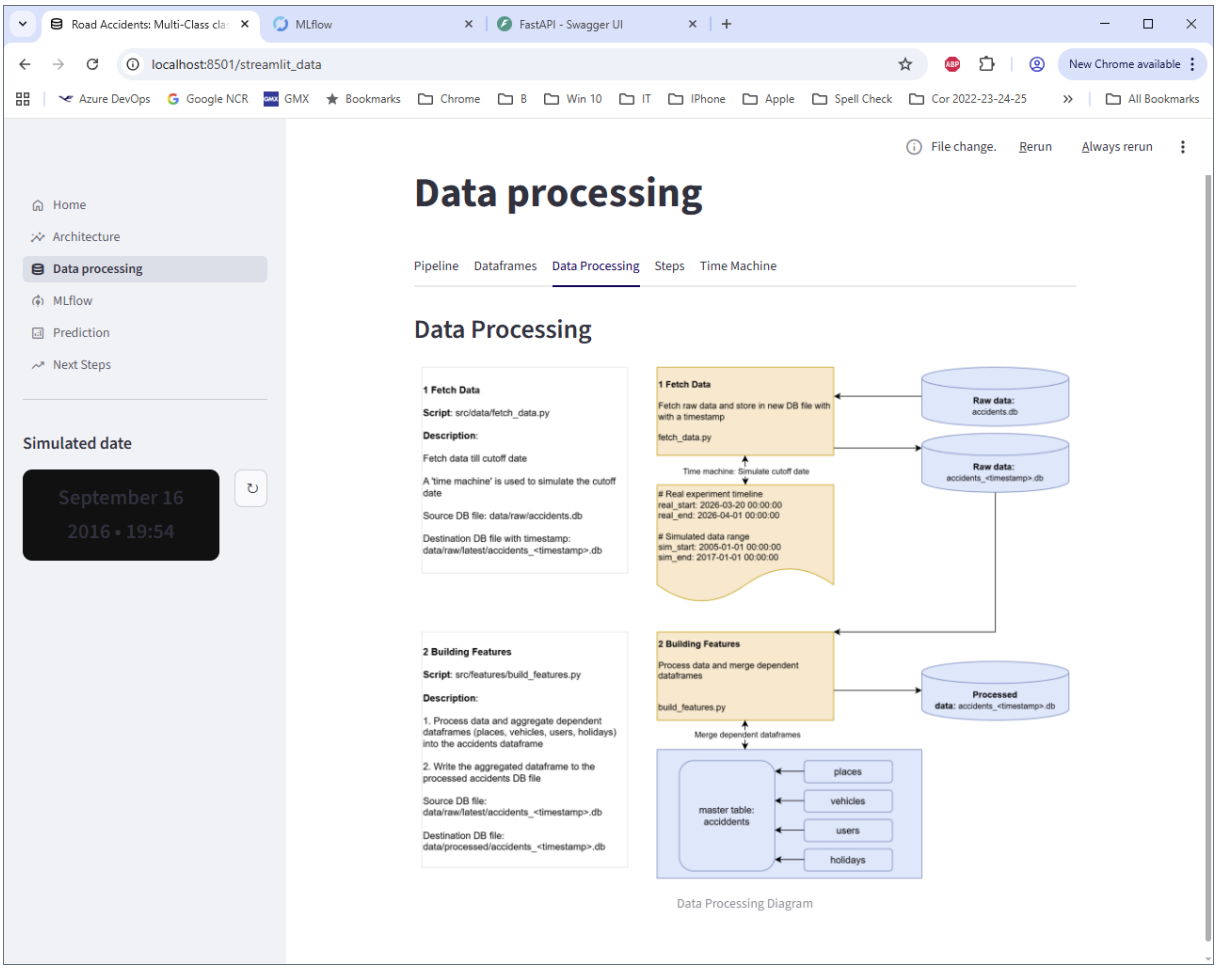
Dataframes
Raw data consist of 5 dataframes:
Accidents (master)
Places
Vehicles
Users
Holidays
master table: accidents
places
vehicles
users
holidays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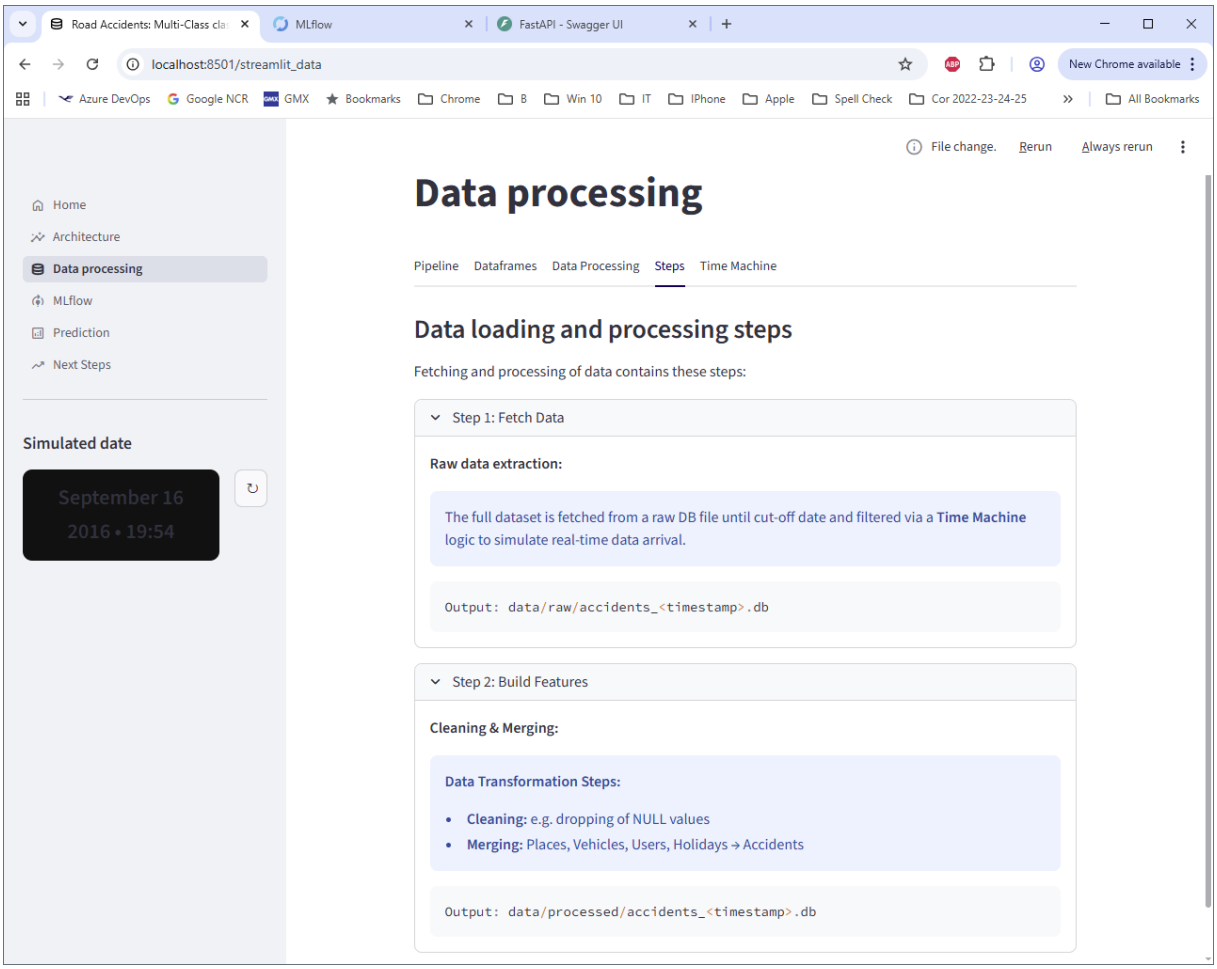
Data Processing
1 Fetch Data
Script: src/data/fetch_data.py
Description:
Fetch data till cutoff date
A 'time machine' is used to simulate the cutoff date
Source DB file: data/raw/accidents.db
Destination DB file with timestamp: data/raw/latest/accidents_<timestamp>.db
1 Fetch Data
Fetch raw data and store in new DB file with with a timestamp
fetch_data.py
Raw data: accidents.db
Raw data: accidents_<timestamp>.db
Time machine: Simulate cutoff date
# Real experiment timeline
real_start: 2026-03-20 00:00:00
real_end: 2026-04-01 00:00:00
# Simulated data range
sim_start: 2005-01-01 00:00:00
sim_end: 2017-01-01 00:00:00
2 Building Features
Script: src/features/build_features.py
Description:
1. Process data and aggregate dependent dataframes (places, vehicles, users, holidays) into the accidents dataframe
2. Write the aggregated dataframe to the processed accidents DB file
Source DB file: data/raw/latest/accidents_<timestamp>.db
Destination DB file: data/processed/accidents_<timestamp>.db
2 Building Features
Process data and merge dependent dataframes
build_features.py
Processed data: accidents_<timestamp>.db
Merge dependent dataframes
master table: acciddents
places
vehicles
users
holidays
Data Processing Diagram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Data loading and processing steps
Fetching and processing of data contains these steps:
Step 1: Fetch Data
Raw data extraction:
The full dataset is fetched from a raw DB file until cut-off date and filtered via a Time Machine logic to simulate real-time data arrival.
Output: data/raw/accidents_<timestamp>.db
Step 2: Build Features
Cleaning & Merging:
Data Transformation Steps:
Cleaning: e.g. dropping of NULL values
Merging: Places, Vehicles, Users, Holidays → Accidents
Output: data/processed/accidents_<timestamp>.db

Road Accidents: Multi-Class cla | MLflow | FastAPI - Swagger UI

localhost:8501/streamlit_data

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New Chrome available

File change. Rerun Always rerun

- Home
- Architecture
- Data processing
- MLflow
- Prediction
- Next Steps

**Simulated date**

September 16
2016 • 19:54

# Data processing

Pipeline | Dataframes | Data Processing | Steps | Time Machine

## The Time Machine Concept

**Goal:** To simulate a moving "Present" within a historical dataset.
It maps the **real-world timeframe** of the experiment to a **simulated timeframe**, ensuring the experiment progresses at the same relative pace.

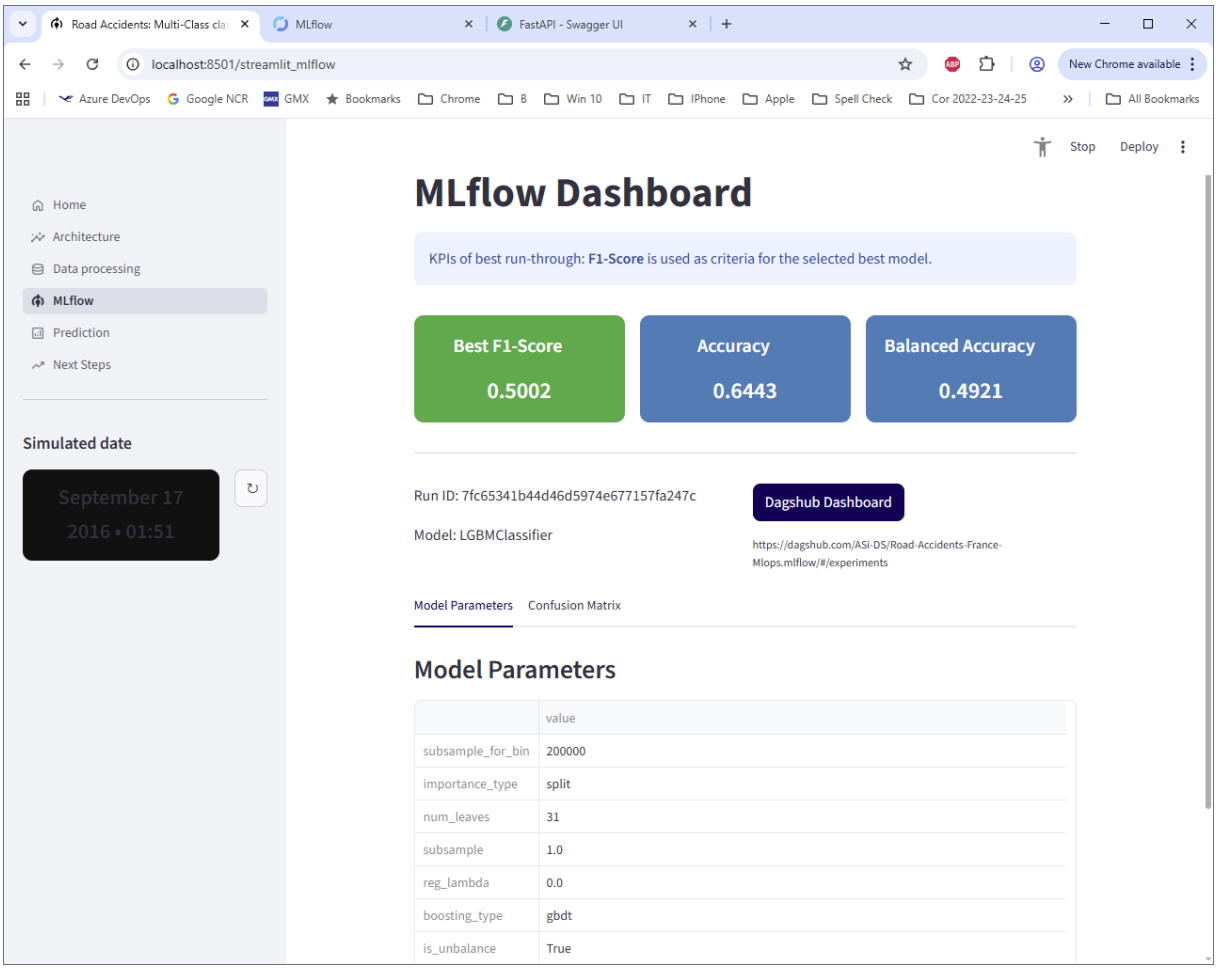
How the Mapping Logic works

- **Normalization (Ratio):** Calculates the progress of the real experiment (e.g., halfway between `real_start` and `real_end` = **0.5**).
- **Linear Mapping:** Applies that same ratio to the simulated range. A ratio of 0.5 returns a timestamp exactly halfway between `sim_start` and `sim_end`.
- **DB Filtering (Cut-off):** The resulting `sim_time` acts as a moving cut-off. We query: `timestamp <= sim_time`.
- **In short:** It answers: *"How much time has passed since I started the script?"* vs. *"What date would that be in my historical data?"*

This concept allows to hide future data that 'hasn't happened yet' in the simulation.

Current Time Machine Configuration:


```
{
  "experiment": {
    "real_start": "2026-03-20 00:00:00",
    "real_end": "2026-04-01 00:00:00",
    "sim_start": "2005-01-01 00:00:00",
```


Road Accidents: Multi-Class cla
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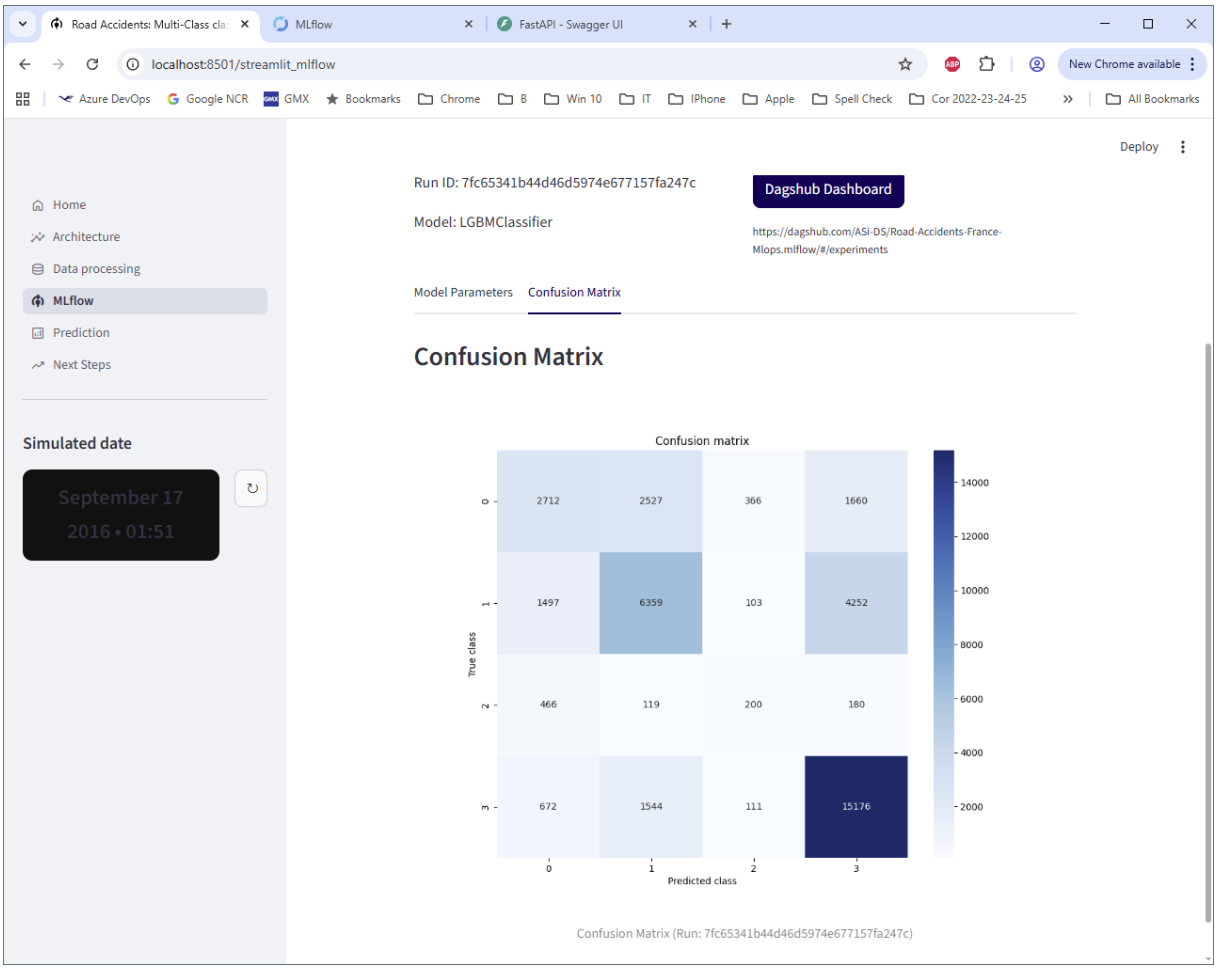
MLflow Dashboard
KPIs of best run-through: F1-Score is used as criteria for the selected best model.
Best F1-Score
0.5002
Accuracy
0.6443
Balanced Accuracy
0.4921
Run ID: 7fc65341b44d46d5974e677157fa247c
Model: LGBMClassifier
Dagshub Dashboard
https://dagshub.com/ASi-DS/Road-Accidents-France-Mlops.mlflow/#/experiments
Model Parameters
Confusion Matrix
Model Parameters
value
subsample_for_bin 200000
importance_type split
num_leaves 31
subsample 1.0
reg_lambda 0.0
boosting_type gbdt
is_unbalance True


Road Accidents: Multi-Class cla
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GMX
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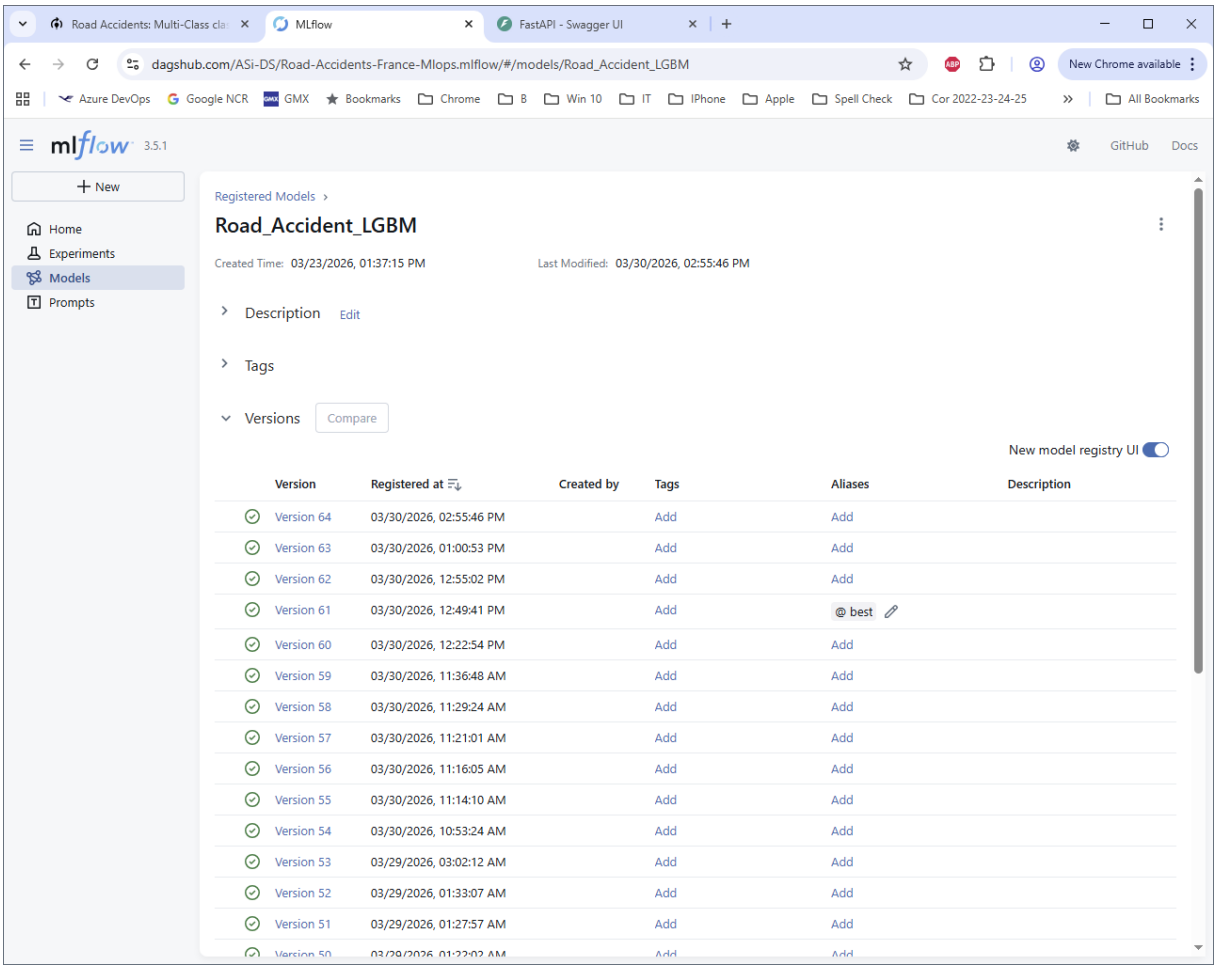
Run ID: 7fc65341b44d46d5974e677157fa247c
Model: LGBMClassifier
Dagshub Dashboard
https://dagshub.com/ASi-DS/Road-Accidents-France-Mlops.mlflow/#/experiments
Model Parameters
Confusion Matrix
Confusion Matrix
Confusion matrix
True class
Predicted class
2712
2527
366
1660
1497
6359
103
4252
466
119
200
180
672
1544
111
15176
0
1
2
3
14000
12000
10000
8000
6000
4000
2000
Confusion Matrix (Run: 7fc65341b44d46d5974e677157fa247c)


Road Accidents: Multi-Class cla
MLflow
FastAPI - Swagger UI
dagshub.com/ASi-DS/Road-Accidents-France-Mlops.mlflow/#/models/Road_Accident_LGBM
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GMX
Bookmarks
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Apple
Spell Check
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mlflow 3.5.1
GitHub
Docs
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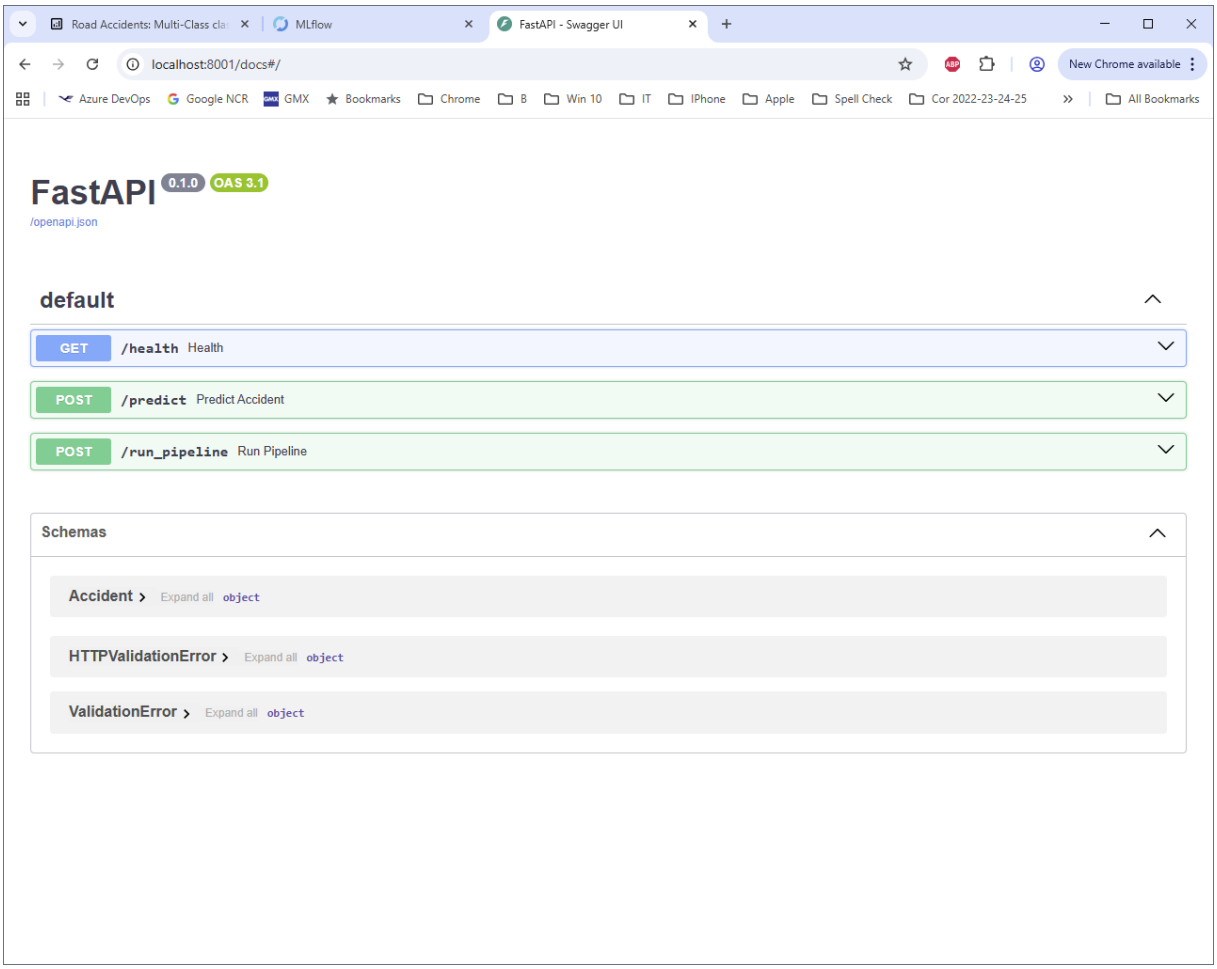
Experiments
Models
Prompts
Registered Models
Road_Accident_LGBM
Created Time: 03/23/2026, 01:37:15 PM
Last Modified: 03/30/2026, 02:55:46 PM
Description
Edit
Tags
Versions
Compare
New model registry UI
Version
Registered at
Created by
Tags
Aliases
Description
Version 64 03/30/2026, 02:55:46 PM Add Add
Version 63 03/30/2026, 01:00:53 PM Add Add
Version 62 03/30/2026, 12:55:02 PM Add Add
Version 61 03/30/2026, 12:49:41 PM Add @ best
Version 60 03/30/2026, 12:22:54 PM Add Add
Version 59 03/30/2026, 11:36:48 AM Add Add
Version 58 03/30/2026, 11:29:24 AM Add Add
Version 57 03/30/2026, 11:21:01 AM Add Add
Version 56 03/30/2026, 11:16:05 AM Add Add
Version 55 03/30/2026, 11:14:10 AM Add Add
Version 54 03/30/2026, 10:53:24 AM Add Add
Version 53 03/29/2026, 03:02:12 AM Add Add
Version 52 03/29/2026, 01:33:07 AM Add Add
Version 51 03/29/2026, 01:27:57 AM Add Add


Road Accidents: Multi-Class cla
MLflow
FastAPI - Swagger UI
localhost:8001/docs#/
New Chrome available
Azure DevOps
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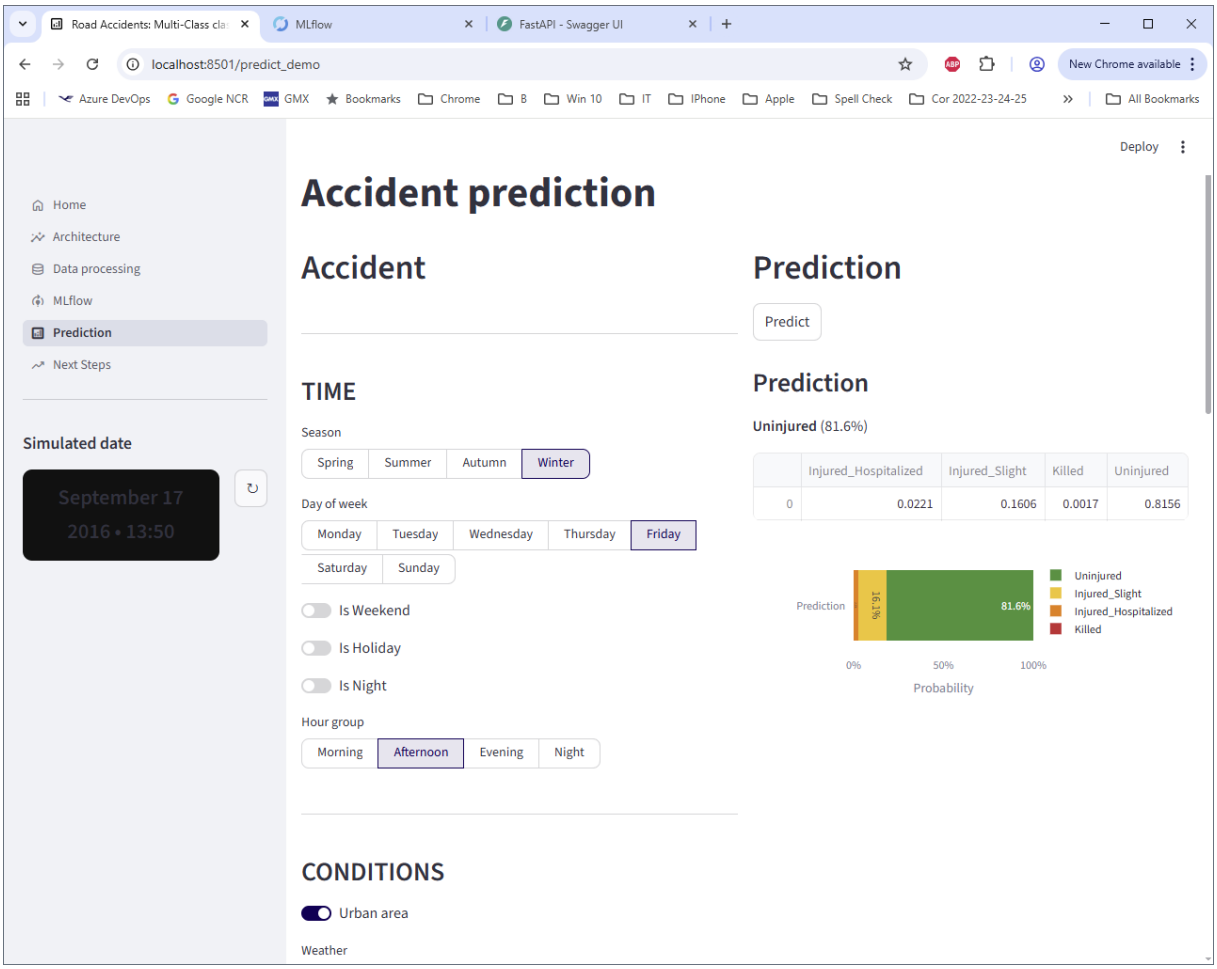
FastAPI
0.1.0
OAS 3.1
/openapi.json
default
GET /health Health
POST /predict Predict Accident
POST /run_pipeline Run Pipeline
Schemas
Accident Expand all object
HTTPValidationError Expand all object
ValidationError Expand all object


Road Accidents: Multi-Class cla
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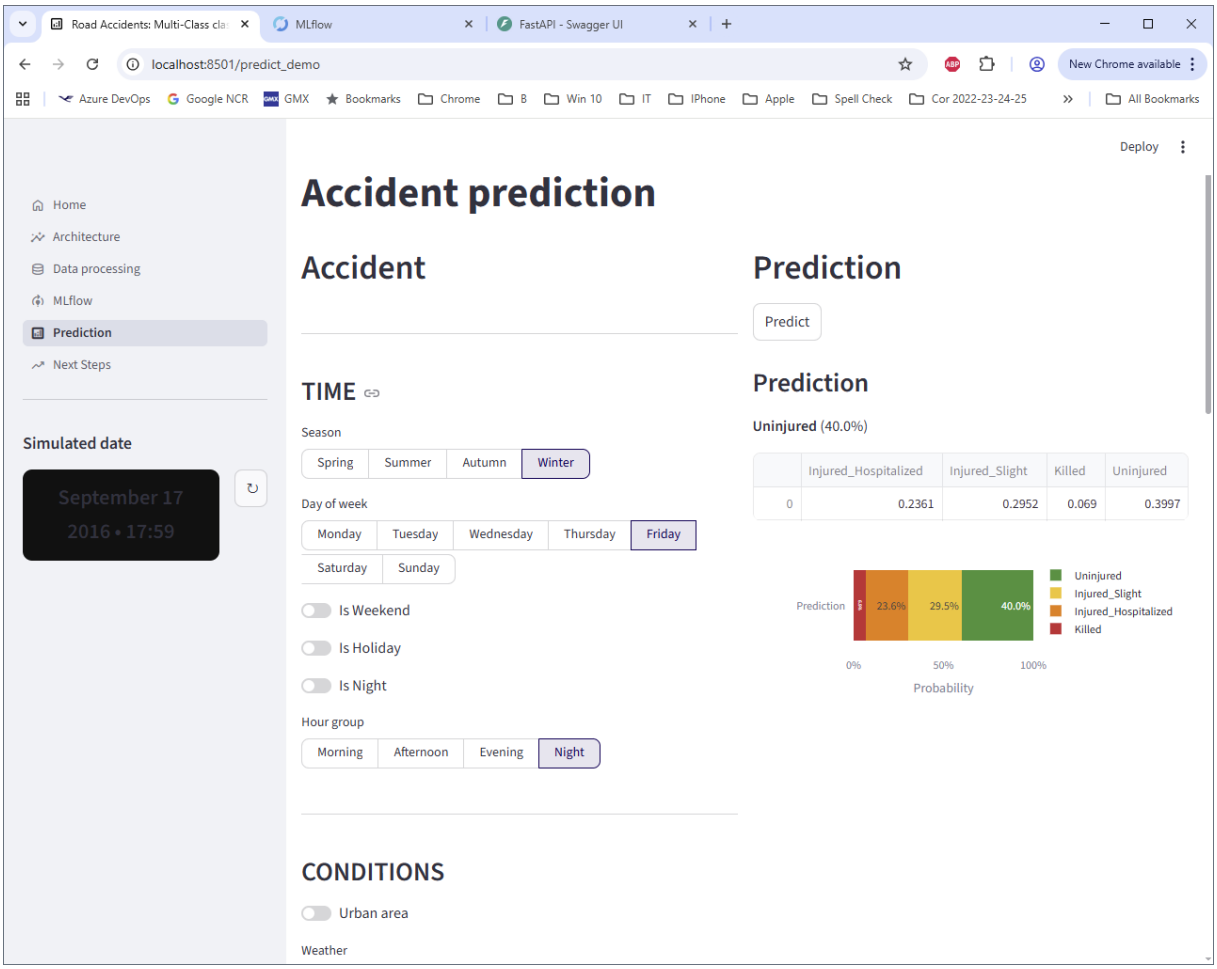
Accident prediction
Accident
TIME
Season
Spring
Summer
Autumn
Winter
Day of week
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Is Weekend
Is Holiday
Is Night
Hour group
Morning
Afternoon
Evening
Night
CONDITIONS
Urban area
Weather
Prediction
Predict
Prediction
Uninjured (81.6%)
Injured_Hospitalized
Injured_Slight
Killed
Uninjured
0
0.0221
0.1606
0.0017
0.8156
Prediction
16.1%
81.6%
Uninjured
Injured_Slight
Injured_Hospitalized
Killed
0%
50%
100%
Probability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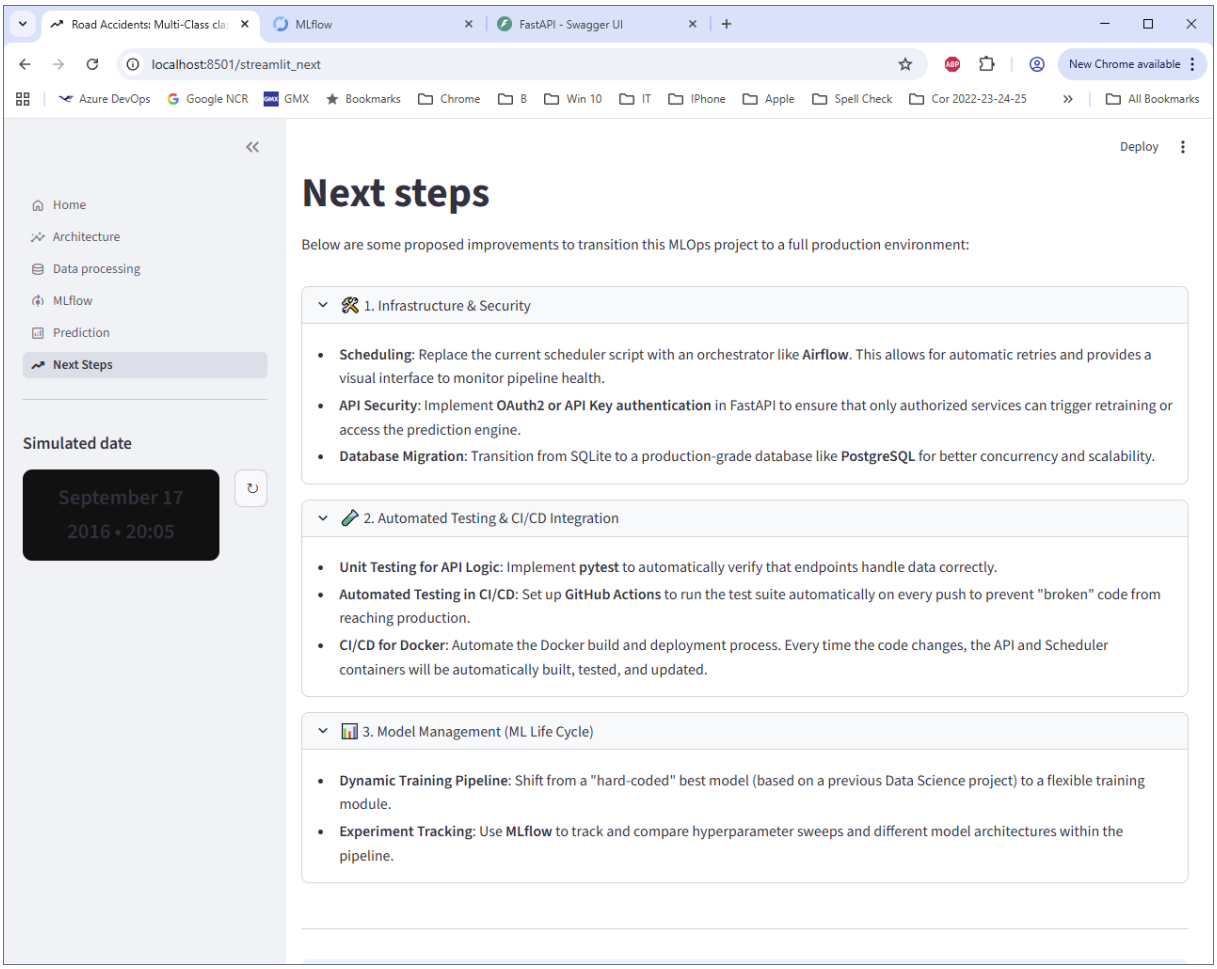
Accident prediction
Accident
TIME
Season
Spring
Summer
Autumn
Winter
Day of week
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Is Weekend
Is Holiday
Is Night
Hour group
Morning
Afternoon
Evening
Night
CONDITIONS
Urban area
Weather
Prediction
Predict
Prediction
Uninjured (40.0%)
Injured_Hospitalized
Injured_Slight
Killed
Uninjured
0
0.2361
0.2952
0.069
0.3997
Prediction
23.6%
29.5%
40.0%
Uninjured
Injured_Slight
Injured_Hospitalized
Killed
0%
50%
100%
Probability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Next steps
Below are some proposed improvements to transition this MLOps project to a full production environment:
1. Infrastructure & Security
Scheduling: Replace the current scheduler script with an orchestrator like Airflow. This allows for automatic retries and provides a visual interface to monitor pipeline health.
API Security: Implement OAuth2 or API Key authentication in FastAPI to ensure that only authorized services can trigger retraining or access the prediction engine.
Database Migration: Transition from SQLite to a production-grade database like PostgreSQL for better concurrency and scalability.
2. Automated Testing & CI/CD Integration
Unit Testing for API Logic: Implement pytest to automatically verify that endpoints handle data correctly.
Automated Testing in CI/CD: Set up GitHub Actions to run the test suite automatically on every push to prevent "broken" code from reaching production.
CI/CD for Docker: Automate the Docker build and deployment process. Every time the code changes, the API and Scheduler containers will be automatically built, tested, and updated.
3. Model Management (ML Life Cycle)
Dynamic Training Pipeline: Shift from a "hard-coded" best model (based on a previous Data Science project) to a flexible training module.
Experiment Tracking: Use MLflow to track and compare hyperparameter sweeps and different model architectures within the pipeline.